epochs or so. Bottom: The 1D Ising model after maximizing rewards (minimizing energy). You can see that all the electrons are clustered together into domains where they are all oriented the same way.

Figure 9.18. Left: A single bar magnet and its magnetic field lines. Recall that a magnet has two magnetic poles, often called North (N) and South (S). Right: Put two bar magnets close together, and their combined magnetic field is a bit more complicated. When we're modeling how the electron spins behave in a 2D or 3D grid, we care about the overall magnetic field generated by the contributations of all the electrons in a neighborhood; we don't need to know what the magnetic field is for each individual electron.

Figure 9.19. The joint action for a pair of electron spins is the outer product between their individual action vectors, which is a 4-element one-hot vector. Rather than using this exact joint action, we can approximate it by taking the average of these two action vectors, resulting in what's called the mean field approximation. For two electrons together, with one spin-up and the other spin-down, the mean field approximation results in reducing this two electron system to a single "virtual" electron with an indeterminate spin of [0.5,0.5].

Figure 9.20. We represent the 2D Ising model as a 2D square grid (i.e., a matrix), but we design it so that there are no boundaries, and the agents that appear to be on a boundary are actually immediately adjacent to the agents on the opposite side of the grid. Thus the 2D grid is really a 2D grid wrapped around the surface of a sphere.

Figure 9.21. This is a randomly initialized 2D Ising model. Each grid square represents an electron. The light-colored grid squares represent electrons oriented with spin-up and the dark squares are spin-down.

Figure 9.22. The top graph is the loss plot for the DQN. The loss doesn't look like it is converging, but we can see that it does indeed learn to minimize the energy of the system (maximize reward) as a whole in the bottom panel.

Figure 9.23. Screenshot from the MAgent multiplayer Gridworld game with two opposing teams of Gridworld agents. The objective is for each team to kill the other.